INFO6105 FINAL PROJECT
PHIUSIIL PHISHING URLS

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INTRODUCTION

SCOPE OF THE STUDY

Phishing attacks are a major cybersecurity threat, using deceptive URLs to trick users into revealing sensitive information

RELEVANCE OF THE STUDY

The project applies a data science lens to explore the PhiUSIIL Phishing URL

Objective

To derive actionable insights for cybersecurity—not model building—but to inform future detection systems

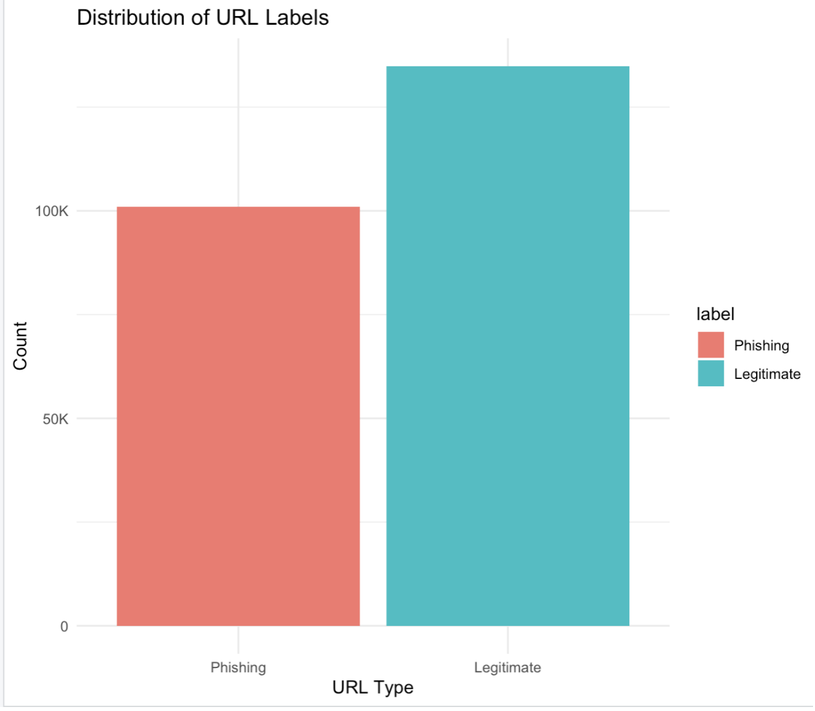
RESEARCH QUESTIONS

WHAT ARE THE KEY DISTINGUISHING FEATURES BETWEEN LEGITIMATE AND PHISHING URL?

HOW DO URL CHARACTERISTICS SUCH AS TLDLEGITIMATEPROB OR URLTITLEMATCHSCORE AFFECT THE LIKELIHOOD OF A MALICIOUS URL?

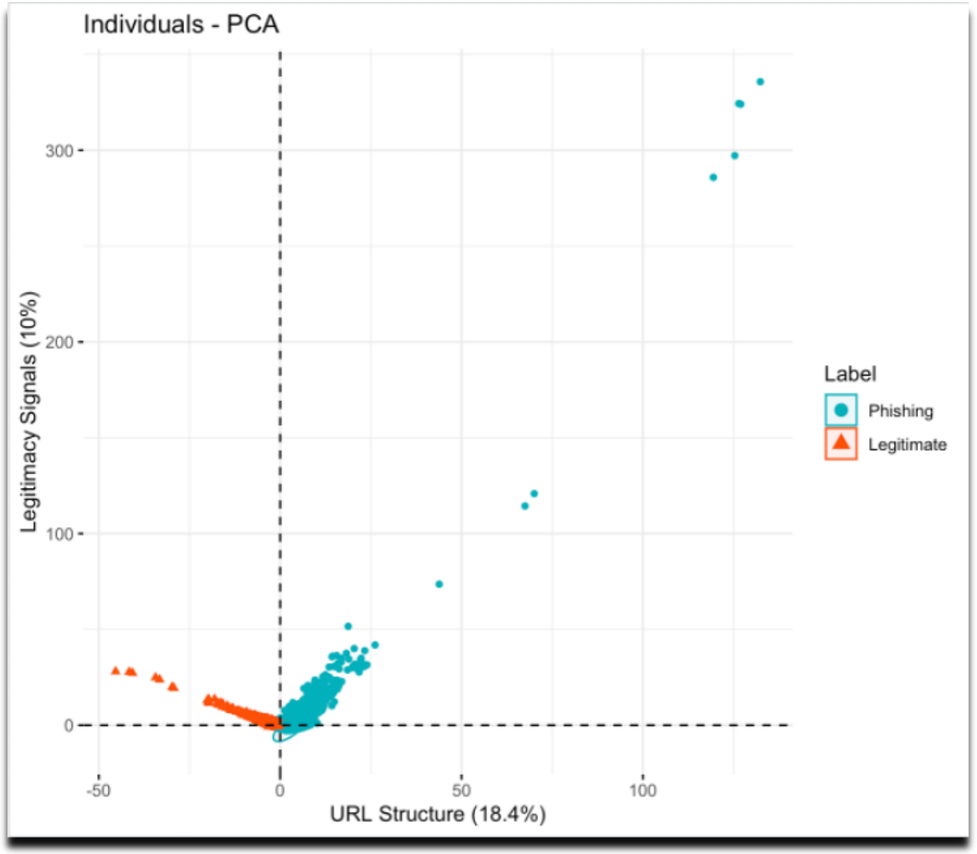
METHODOLOGY

- Data Processing and Exploratory Data Analysis (EDA)
- Feature Selection



Feature	Description	Key Observation	Interpretation
Label	URL type (Phishing or Legitimate)	Legitimate: 134,850 (56.8%) Phishing: 100,945 (43.2%)	Slight class imbalance
TLDLegitimateProb	Probability that a URL's TLD (Top-Level Domain) is associated with legitimate sites	Phishing URLs are concentrated near 0.0 Legitimate URLs peak near 0.5+	Legitimate domains tend to use trusted TLDs
URLTitleMatchScore	Score indicating how well the page title matches the domain	Legitimate URLs skew toward higher scores	Phishing sites often use mismatched or deceptive titles
CharContinuationRate	Measures character pattern continuity	Legitimate URLs cluster around 1.0 Phishing URLs have more variation	Phishing URLs tend to have unnatural patterns (e.g., random characters)
Missing Values	Presence of NA in columns	None detected across dataset	Dataset is clean and ready for modeling

METHODOLOGY



Question 1: What are the key distinguishing features between legitimate and phishing URL?

Process:

- Welch's t-tests on selected features: TLDLegitimateProb, URLTitleMatchScore, CharContinuationRate
- Boxplots for visual comparison
- Effect size calculation using Cohen's d
- Principal Component Analysis (PCA) to uncover patterns in high-dimensional data

Key Findings:

- URLTitleMatchScore: Significantly lower in phishing URLs (Mean = 21.2) vs. legitimate (Mean = 75.3); Cohen's d = -1.29 (large effect)
- CharContinuationRate: Lower in phishing (Mean = 0.73) than legitimate (Mean = 0.93); Cohen's d = -1.07 (large effect)
- TLDLegitimateProb: Slightly lower in phishing (Mean = 0.23) than legitimate (Mean = 0.28); Cohen's d = -0.20 (negligible effect)
- In PCA, URLTitleMatchScore and CharContinuationRate are key distinguishing features with strong statistical and practical significance in separating phishing from legitimate URLs.

METHODOLOGY

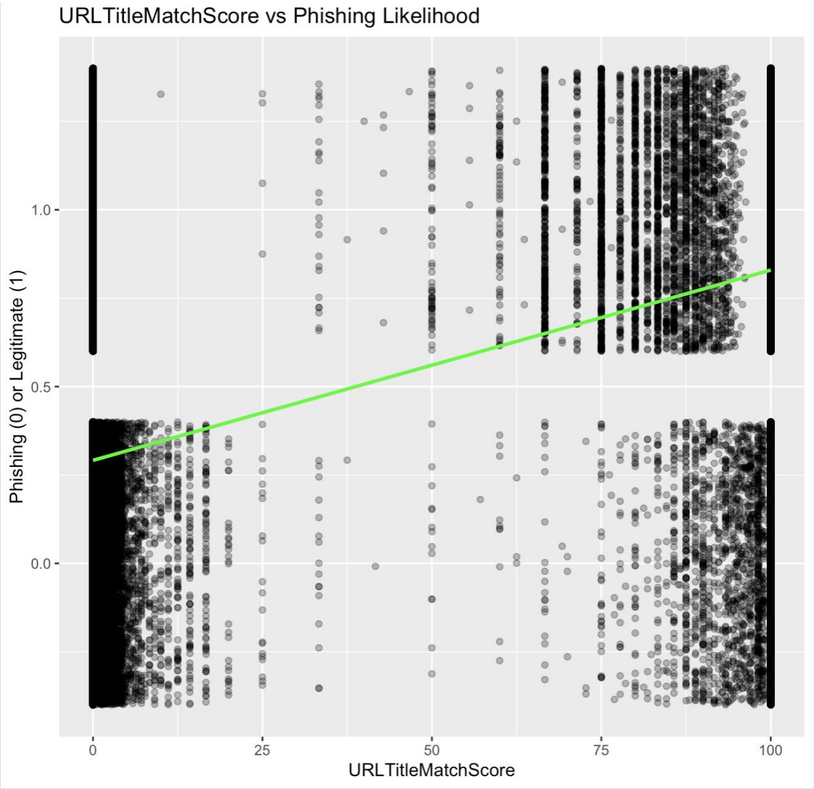
Question 2: How do URL characteristics such as TLDLegitimateProb or URLTitleMatchScore affect the likelihood of a malicious URL?

Process:

- Correlation analysis with label_numeric (0 = phishing, 1 = legitimate)
- Scatterplots with linear trend lines
- ANOVA tests for statistical significance

Key Findings:

- Positive correlation of URLTitleMatchScore (+0.61) and CharContinuationRate (+0.50) with legitimacy
- ANOVA results confirm all three features differ significantly between phishing and legitimate URLs ($p < 0.001$)



CONCLUSION

- Key Features Matter: Statistical tests and PCA revealed that features like URLTitleMatchScore and CharContinuationRate significantly differ between phishing and legitimate URLs, offering strong discriminatory power.

- Legitimacy Signals Are Telling: High URLTitleMatchScore and TLDLegitimateProb values are positively associated with legitimate URLs, as confirmed by correlation and scatterplot analyses.

- Data Science Adds Clarity: Through EDA, hypothesis testing, PCA, and visualizations in R, we gained valuable insights into phishing behaviors — laying a foundation for future detection systems.

THANK YOU